

DENIS MUTHURI

COMPUTER SCIENCE STUDENT | SOFTWARE DEVELOPMENT | AI

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PROFESSIONAL PROFILE

Motivated Bachelor of Science in Computer Science student at Dedan Kimathi University of Technology with practical experience in software development, artificial intelligence, web technologies, databases, networking and computer systems. Experienced in team-based projects involving full-stack applications, AI systems, NLP, Android development and knowledge-based reasoning. Seeking an external attachment to apply technical skills in a professional environment while gaining practical industry experience. Available for an 8–12 week attachment placement.

TECHNICAL SKILLS

Programming: Python, Java, C, C++, JavaScript, PHP

Web & Backend: HTML, CSS, JavaScript, Node.js, Express.js, REST APIs

Databases: PostgreSQL, SQL

AI & Computing: Artificial Intelligence, Knowledge Representation & Reasoning, NLP, CFG, PDA, Turing Machines

Tools: Git, GitHub, Streamlit, Android Studio, JFLAP, JFlex, IntelliJ IDEA, Cisco Packet Tracer

Other: JWT, bcrypt, database design, debugging, testing, technical documentation, teamwork

SELECTED PROJECTS

Church Financial Management System (CFMS) — Full-Stack Web Application

- Developed a system to digitize church contributions, expenditures, members, groups and projects. Built backend services with Node.js/Express and PostgreSQL; implemented JWT/bcrypt authentication and worked on REST APIs, database relationships, validation and transaction recording.

Intelligent Healthcare Diagnostic Assistant — AI Capstone

- Developed a multi-module AI-based healthcare diagnostic assistant using Python and Streamlit. Applied AI and knowledge-based reasoning concepts and collaborated on integration, testing and demonstration.

Smart Grammar-Based FAQ Assistant — NLP

- Developed a bilingual voice/text FAQ assistant using NLP and Context-Free Grammar concepts. Worked on query processing, structured responses and WhatsApp deployment for a university information-access scenario.

Reasoning Systems — AI/KRR

- Developed an e-commerce recommendation system and a smart-home production-rule system, applying Knowledge Representation and Reasoning concepts to practical decision-making scenarios.

Miami Supermarket Pay-Point App — Android

- Developed a supermarket pay-point application in Android Studio as part of a team project, working on application functionality and user interaction.

TECHNICAL / ACADEMIC EXPERIENCE

Internal Technical Attachment — Dedan Kimathi University of Technology

- Gained practical exposure to PC assembly, motherboards, RAM, chipsets, CMOS, BIOS/UEFI and POST fault diagnosis.
- Worked with IP/MAC addressing, VLANs, subnetting, network devices and crossover cables using Cisco Packet Tracer.
- Gained exposure to Arduino and basic hardware/software interaction, technical documentation and troubleshooting.

EDUCATION

Dedan Kimathi University of Technology (DeKUT) — Bachelor of Science in Computer Science | Expected Graduation: April 2028

Nkubu High School — Meru — KCSE, 2020–2023 | Grade: B+

CERTIFICATIONS

- Computer Packages Certificate
- Fundamentals of Cybersecurity Certificate

ADDITIONAL PROJECTS

Java Sales Invoice Generator: Java application for customer/product information, taxes, discounts and invoice calculations.

SHIF CFG/PDA System: Designed Context-Free Grammars and Pushdown Automata to model SHIF claims messages.

Course Registration System: Frontend application developed using HTML, CSS and JavaScript with team task tracking.

CORE STRENGTHS

Problem solving • Analytical thinking • Software development • Technical troubleshooting • Team collaboration • Research • Documentation • Willingness to learn

AVAILABILITY

Available for an 8–12 week external attachment placement.

REFEREES

Available upon request.